

Methodological integrity document

Simulation 3 explained: what is evidenced, what is hypothetical

DCEN + EducationLibre + European ENR Coalition

- Part A: facts supported by verifiable sources
- Part B: the modelling and its arbitrary figures
 - Why this distinction is essential
- How to read the simulation results honestly

This document rigorously separates documented facts from modelling assumptions.
The simulation figures are NOT predictions or proven results.

Foreword: why this document exists

Simulation 3 produces figures: 84% renewable energy, +40 points of autonomous governance, 85% retraining of fossil-fuel workers. These figures have the appearance of scientific results. They are not.

They are outputs of an illustrative model built with hand-chosen formulas, whose coefficients were calibrated to make the system internally coherent. Presenting these figures as proof would be dishonest and would weaken the project against any serious criticism.

This document therefore explicitly separates two things: on one side the real facts, supported by verifiable academic and institutional sources (Part A); on the other side the modelling and its arbitrary figures (Part B). The reader can thus know, for each statement, whether it is a documented fact or a working hypothesis.

EVIDENCED FACT	Statement supported by a verifiable, cited source
MODELLING	Assumption or figure produced by the model, with no predictive value

Part A - Facts supported by sources

All elements in this part are documented by verifiable sources, cited at the foot of each section. These are the real foundations on which the DCEN model rests.

A.1 Wealth concentration and tax evasion (Zucman)

EVIDENCED
FACT

About 8% of global financial wealth is in tax havens

In The Hidden Wealth of Nations (2015), Gabriel Zucman estimates that around \$7.6 trillion - roughly 8% of household financial wealth - was held in offshore centres in 2014 (Switzerland, Luxembourg, Cayman Islands, etc.), causing tax losses of around \$200 billion per year for states.

EVIDENCED
FACT

Billionaires pay an effective tax rate of about 0.3% of their wealth

In his report for the Brazilian G20 presidency (2024), Zucman documents that the roughly 3,000 global billionaires pay an effective rate equivalent to 0.2-0.3% of their wealth. His proposal for a 2% minimum tax on wealth above \$1 billion would raise between \$200 and \$250 billion per year.

EVIDENCED
FACT

This proposal has real political support

The Zucman proposal was brought to the G20 and supported by the Brazilian (Lula) and French (Macron) presidents, as well as ministers from South Africa, Spain, France and Germany. In February 2025, the French National Assembly adopted a variant: a minimum tax of 2% on wealth above 100 million euros.

Sources: G. Zucman, The Hidden Wealth of Nations (2015); G. Zucman, A blueprint for a coordinated minimum effective taxation standard for ultra-high-net-worth individuals, EU Tax Observatory / G20 report (2024); Wikipedia, Global minimum tax on billionaires.

A.2 The four-day week / working-time reduction (Iceland)

EVIDENCED FACT

The Icelandic trials involved about 2,500 workers (1% of the workforce)

Between 2015 and 2019, trials run by Reykjavik City Council and the Icelandic government moved workers from 40h to 35-36h per week with no loss of pay, across more than 60 workplaces (schools, hospitals, social services).

EVIDENCED FACT

Productivity stayed the same or improved, wellbeing increased

Researchers (think tanks Autonomy and Alda) concluded it was an 'overwhelming success': productivity remained the same or increased in most workplaces, while workers' wellbeing increased markedly (stress, burnout, work-life balance).

EVIDENCED FACT

The trials led to permanent changes

Following the trials, Icelandic unions negotiated hour reductions for tens of thousands of members. By 2021, about 86% of Icelandic workers were working less or had the right to reduce their hours. In 2023, the Icelandic economy grew by 5%, the second-highest in Europe according to the IMF.

Sources: G. Haraldsson, J. Kellam (Autonomy & Alda), Going Public: Iceland's journey to a shorter working week (2021); CNN Business (Oct. 2024); Washington Post (Jul. 2021); World Economic Forum (2022); IMF World Economic Outlook (2024).

A.3 Cooperatives at scale (Mondragon)

EVIDENCED FACT

Mondragon employs about 70,000 people in 2024

Founded in 1956 by the priest Jose Maria Arizmendiarieta with his students, the Mondragon Corporation employed 70,085 people in 2024, with revenue of 11.2 billion euros. It is the world's largest worker-cooperative federation and the largest employer in the Basque Country.

EVIDENCED FACT

A historically high cooperative survival rate

Of the 103 cooperatives created between 1956 and 1986, only three closed - a survival rate of about 97% over three decades. Mondragon owns its own bank (Laboral Kutxa), its university (4,000 students) and its social-protection system.

EVIDENCED FACT

Education was at the heart of the project from the start

The project began in 1943 with a technical school, before the first industrial cooperative in 1956. Training and education have always been central - a point directly relevant to the articulation between cooperatives and P2P education in the DCEN model.

Sources: Wikipedia / Mondragon Corporation Annual Report 2024; Corporate Rebels (2024); Participedia, Mondragon Co-operative Corporation; Mondragon Unibertsitatea.

A.4 The feasibility of a global tax agreement (2021 OECD precedent)

EVIDENCED FACT

136 countries signed the global minimum tax agreement in October 2021

On 8 October 2021, the OECD announced that 136 countries (out of 140 inclusive-framework members) had reached an agreement imposing a 15% minimum rate on large multinationals. Signatories represented over 90% of global GDP, including jurisdictions regarded as tax havens (Cayman Islands, Gibraltar).

EVIDENCED FACT

This agreement shows global fiscal coordination is possible

The agreement is explicitly inspired by the logic of a global tax floor - exactly the mechanism the DCEN model proposes for wealth. It is the precedent that makes the hypothesis of a coordinated fiscal blockchain credible (but not guaranteed).

EVIDENCED FACT

But this agreement remains fragile and contested

Important for honesty: in 2025, the Trump administration criticised the 2021 agreement, threatening retaliatory taxes, which forced a renegotiation exempting certain US companies. This shows that a global agreement, even when signed, is never definitively secured - a real limitation for the DCEN model.

Sources: OECD, Statement on a Two-Pillar Solution (Oct. 2021); BDO Global Tax (2021); ICIJ (Oct. 2021); Library of Congress (Jul. 2021); Reuters / Deccan Herald (2025, renegotiation).

A.5 Intrinsic motivation and non-monetary credits

EVIDENCED FACT

Financially rewarding an activity can destroy intrinsic motivation

Deci & Ryan's Self-Determination Theory (1985) documents the overjustification effect: introducing an extrinsic reward (money) into an activity already intrinsically motivated (the pleasure of learning, of sharing) can reduce intrinsic motivation. This is the scientific basis for the decision NOT to make EducationLibre credits convertible into money.

EVIDENCED FACT

The non-monetary contribution model works at scale

Wikipedia empirically demonstrates that a voluntary, unpaid contribution system can produce the largest knowledge base in human history. This is not proof that EducationLibre will work, but a demonstration that the principle of non-monetary reciprocity is viable at scale.

Sources: E. Deci & R. Ryan, Intrinsic Motivation and Self-Determination in Human Behavior (1985); Wikimedia Foundation (funding and contribution model).

Part B - The modelling and its arbitrary figures

This part explains how the simulation figures were produced, and why they must never be presented as proof or predictions.

B.1 How the figures were fabricated

MODELLING

The indicators are outputs of hand-chosen formulas

Each percentage (84% renewables, 78% citizen autonomy, etc.) is the result of a weighted formula of the type: $\text{indicator} = a \cdot \text{parameter1} + b \cdot \text{parameter2} + \dots$, where the coefficients $a, b, \dots$ were chosen to produce results coherent with intuition. These coefficients rest on no empirical data.

MODELLING

Comparisons between simulations are constructions, not measurements

When the document states '+40 points of governance-without-experts between sim.2 and sim.3', this gain is entirely determined by how the formulas were written. It reflects a design assumption ('EducationLibre should greatly improve autonomy'), not an observation.

MODELLING

The 'phase factor' is an arbitrary multiplication

The temporal phases (bootstrapping, growth, maturity) apply multipliers (0.6 / 1.0 / 1.35) chosen to represent a plausible ramp-up. These values were not derived from real transition studies.

B.2 What the modelling can and cannot say

The modelling CAN	The modelling CANNOT
Show the internal coherence of a system of ideas	Predict what will actually happen
Illustrate relationships between variables (if X rises, Y tends to...)	Provide reliable absolute values (84%, 200 members...)
Help identify tensions and critical thresholds	Replace real-world field experimentation
Serve as a tool for discussion and clarification	Be cited as scientific proof
Make design assumptions explicit	Validate that the DCEN model will work

B.3 The Simulation 3 figures, honestly requalified

The table below takes the main figures from simulation 3 and indicates, for each, what it really represents.

Figure shown	Real status	Honest interpretation
84% renewables in mix	Assumption	Illustrative target value, not a forecast. No source guarantees this level.
+40 pts governance without experts	Construction	Reflects the assumption that P2P education improves autonomy - plausible direction, invented magnitude.
85% fossil retraining	Assumption	Optimistic. No study measures such a rate for this type of scheme.
15 countries = critical mass	Analogy	Inspired by the OECD agreement (136 countries). The threshold of 15 is a choice, not a calculation.
8% of fund to infrastructure	Chosen parameter	Model input value, set by the user, not derived from data.
850+ sub-subjects	Illustration	Imagined order of magnitude to give a sense of scale, with no empirical basis.

In short: the figures in Part B are illustrations of a line of reasoning, not results. They help to think through the system, not to prove that it will work.

Synthesis: how to use this work honestly

What can be asserted without risk

- Wealth inequality and offshore tax evasion are massive and documented (Zucman).
- Working-time reduction can preserve productivity and improve wellbeing (Iceland).
- Cooperatives can operate at scale and endure (Mondragon, 70,000 jobs).
- Global fiscal coordination is possible but fragile (OECD 2021, then 2025 renegotiation).
- Intrinsic motivation justifies not monetising educational credits (Deci & Ryan).

What must always be specified

- The DCEN model combines these elements in a new way - this combination has never been tested.
- The simulation figures are illustrative, never predictive.
- No source validates that the full assembly will work as the model suggests.
- EducationLibre is a project under development, not a proven system.
- The only way to know if it works is real small-scale experimentation (e.g. a pilot cooperative).

The honest framing for presenting the project

The DCEN model rests on solid, documented facts concerning inequality, working time, cooperatives and fiscal coordination. It proposes a new way of combining these proven elements. The simulations that accompany it are thinking tools that illustrate relationships and tensions - they do not predict the future and do not prove that the assembly will work. The only possible proof would come from real experimentation, for example a first pilot cooperative whose results would be measured and published.

FULL SOURCES - Part A (evidenced facts): G. Zucman, *The Hidden Wealth of Nations* (University of Chicago Press, 2015); G. Zucman, *A blueprint for a coordinated minimum effective taxation standard for UHNWI*, EU Tax Observatory / G20 report (2024); Autonomy & Alda, *Going Public: Iceland's journey to a shorter working week* (2021); Mondragon Corporation, *Annual Report 2024*; OECD, *Statement on a Two-Pillar Solution to Address the Tax Challenges* (8 Oct. 2021); E. Deci & R. Ryan, *Intrinsic Motivation and Self-Determination in Human Behavior* (Plenum, 1985); Wikimedia Foundation. Part B (modelling): formulas and coefficients built for simulations 1, 2 and 3 of this work - with no empirical value. EducationLibre: github.com/nadnone/EducationLibre (CC BY 4.0).